Checkpoint C - Quality and Residual Risk

QuantRule

2026-11-12

Risk map

Risk	Control	Effort	Residual
Look-ahead leaks into a rule	availability bars + causality tests per event	high	a new reading may repeat a past mistake
Non-determinism across runs	fixed accumulation order, sorted merges, canonical digest	high	cross-CPU equality is unproven
A run silently loses an artifact	publish only after every hash verifies	medium	a store failure mid-verify is untested in production
A change breaks recorded evidence	versioned contracts, immutable runs	medium	a digest definition change invalidates old bundles
Wrong input accepted as valid	strict config validation, cross-config checks	medium	a valid config can still be a bad hypothesis

At least one control is non-test: contract versioning and the review rule.

One behaviour, with evidence

Risk: look-ahead. **Behaviour tested:** an event appears only at or after the bar on which it became knowable. **Expected:** appending later bars does not change any earlier published value. **Observed:** the prefix and causality tests pass on the current model suite.

Command: `cargo test -p quantrule-model` (causality and prefix tests).

Limit

Passing causality tests prove the tested readings are causal. They do not prove that a strategy written on top of them is - the strategy is a separate rule and can still read a level before it is confirmed.

Three decisions

- **Review.** The review kept the model's numeric output and rejected adding labels at the service boundary; labels are a client mapping, so two clients cannot drift.

- **AI use.** AI drafted the architecture documents and the issue tree. Every claim in them was checked against the code or the gate; cross-machine determinism is explicitly not claimed as observed.
- **Residual risk.** The determinism claim is bounded to one machine until a second one runs the verification procedure. Next: add the scale-invariance test before the client ships.

Claim -> Evidence -> Limit -> Decision: the evidence supports the causality claim for the tested readings, not for a strategy built on them.